

Foundational Validity Audit of Shared-Exclusions PID

Exact semantics, compatibility firewall, adversarial counterexamples, and a bounded executable audit

pid-rs project

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Abstract

This report audits the categorical shared-exclusions partial information decomposition (SxPID) of Makkeh, Gutknecht, and Wibral from its primitive event semantics. We rederive its lattice order, informative-misinformative split, Möbius inversion, and mutual-information reconstruction; distinguish proved properties from constitutive semantic choices; and give exact finite counterexamples to stronger interpretations. We also reproduce the 2026 Lyu–Clark–Raviv relation witness. A standard-library exact-rational checker and a separate Rust regression show that their stipulated coordinate recoverability descriptors are not SxPID atoms. A kernel-checked Lean theorem then isolates the missing transfer premise: an atom assignment must factor through those descriptors before their collision can imply a reconstruction impossibility for that assignment. No fatal algebraic contradiction is found in the native shared-exclusions construction. The valid conclusion is narrower: SxPID is a coherent signed, local, event-logical decomposition under an explicit semantic and statistical contract, not a universal ontology of informational parts. Natural logarithms and nats are used unless bits are stated explicitly.

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1 Executive decision

The categorical shared-exclusions construction is mathematically valid in the following precise sense.

1. Its keyed source event is well-defined for every supported realization.
2. Its local cumulative is an ordinary likelihood-ratio information value for that event.
3. The antichain order, event containments, and Möbius inversion give an exact signed decomposition.
4. The greatest-node cumulative is local mutual information; averaging the atoms reconstructs joint source–target mutual information.
5. Informative and misinformative component atoms are nonnegative. Net atoms need not be.
6. Source permutation symmetry, coordinatewise bijective re-encoding invariance, self-redundancy, and the target chain rule follow from the definition.

7. The current Rust implementation reproduces the exact finite witnesses in this report, including the adversarial three-source relation witness in Section 8.

No fatal algebraic contradiction was found in the Makkeh–Gutknecht–Wibral construction or in the core theorems audited here. This does *not* show that shared exclusions is the unique correct PID. The following stronger readings are false or unsupported:

- the disjunction event is uniquely forced by Shannon theory;
- an atom is necessarily a nonnegative set-sized part, a common deterministic signal, a common target feature, a causal contribution, or a dependence strength;
- independent sources must have zero Sx redundancy;
- signed net atoms obey a general data-processing inequality;
- pairwise source–target marginals identify the decomposition;
- source and target roles are interchangeable;
- the changing keyed event is one fixed auxiliary random variable; or
- an observational PID atom establishes mechanism, responsibility, safety, deception, or decision authority.

Shared exclusions is a coherent, signed, local, event-logical PID under an explicit semantic contract. It is not a universal ontology of informational parts.

2 Evidence discipline and audit protocol

Every material result is classified as follows.

Label	Meaning
[P]	Published theorem or definition in a primary source.
[R]	Independent exact rederivation in this audit.
[X]	Exact finite counterexample or exact finite witness.
[B]	Bounded exhaustive or seeded computation; evidence, not a universal proof.
[E]	Observation from the current <code>pid-rs</code> executable path.
[O]	Open question or unproved extension.

The protocol was to freeze the native primitive; derive the lattice and reconstruction from events; separate semantics from numerical theorems; seek minimal exact counterexamples; enumerate bounded empirical laws with rational arithmetic; run independent executable witnesses; and apply the compatibility firewall in Section 3. Each conclusion was reviewed through five lenses: definitional compatibility, theorem scope, executable refinement, numerical/statistical implications, and adversarial counterexamples.

2.1 Five-lens result ledger

Result	Definition	Theorem and assumptions	Executable refinement	Numerical/statistical implication	Adversarial finding
F1 Keyed event	Native	Valid for supported finite keys	Rust scans exact equality events	Rare events produce large logs	Unsupported keys have no local occurrence
F2 Lattice/inversion	Native	Finite-poset inversion is exact	Two-four source lattices implemented	Cancellation amplifies rounding	Labels do not prove reconstruction
F3 Component signs	Native	Applies separately to $+$ and $-$, not net	Exact bounded searches agree	Interval signs needed near zero	Exact negative net atoms exist
F4 Averaging	Native	Linearity and top-node self-redundancy	Exact witnesses reconstruct MI	Average is not MI of one fixed keyed event	Relation witness still reconstructs
F5 REI/target chain	Native	Coordinatewise bijections; conditional chain law	Relabeling/lattice tests	Identities hold only within numerical error	Mixing/coarsening are outside REI
F6 Identity/LP	Comparison properties	Sx rejects both generally	COPY and target-copy witnesses	Signed inference is mandatory	Independent sources can have positive redundancy
F7 Disjunction	Native postulate	Logical consequence is valid; interpretation is chosen	Code evaluates that choice	Another semantics is another estimand	No uniqueness theorem follows
F8 Common signal	Stronger incompatible semantics	Not implied by Sx	No common-variable object is computed	Do not call Sx Gács-Körner information	Independent COPY is a counterexample
F9 Role/processing	Native values; stronger DPI incompatible	Direction is built in	Witnesses executable	Preprocessing changes the estimand	Exact role-swap/coarsening failures
F10 Identification	Full joint law required	Population functional is identified by full finite law	Plug-in uses empirical joint PMF	Sparse high-cardinality estimation is hard	Same pairwise marginals, different Sx
F11 Lyu-Clark-Raviv	Their atoms are not Sx atoms	Descriptor theorem is internal; universal wording overbroad	Regression rejects substitution	Relations change Sx atoms	Eight of eighteen atoms differ exactly
F12 Interpretation	Advisory statistic only	No causal or authority theorem	Reports require estimand identity	UQ/selection are separate	Observationally equivalent causal systems defeat attribution

3 Compatibility firewall

No external theorem is transferred to shared exclusions until its objects and assumptions match Sections 4–4.2.

Source or concept	Compatibility	Permitted use	Forbidden transplant
Makkeh–Gutknecht–Wibral (2021)	Native	Definition, cumulatives, component split, target chain, differentiability	Extending finite-discrete or interior-domain theorems beyond their premises
Gutknecht–Wibral–Makkeh (2021)	Native conceptual foundation	Parthood and formal-logic indexing	Calling constitutive principles uniquely forced Shannon theorems
Matthias–Makkeh–Wibral–Gutknecht (2025)	Compatible abstract theorem	LP–TCR–REI and LP–ID–REI incompatibilities	Calling either a defect unique to SxPID or an algebraic inconsistency of Sx
Lyu–Clark–Raviv (2026), Definition 6/Lemma 4	Incompatible atom semantics	Coordinate-recoverability descriptors and relation loss	Calling $H(U_\alpha)$ an SxPID atom or transferring their impossibility theorem without the missing descriptor premise
Williams–Beer identity/local positivity	Comparison only	Report what Sx accepts or rejects	Redefining Sx after a violation
Dependency coloring	Estimator layer only	Concentration under proved mutual within-color independence	Calling it a “colored PID” measure or attributing it to Wibral
pid-rs categorical kernel	Executable approximation	Regression and refinement evidence	Treating tests as proof of semantics or exact-real floating point
Project support-change theorem	Native averaged functional, project theorem	Closed-simplex continuity on a fixed alphabet	Attributing it to the 2021 paper or applying it to disappearing pointwise keys

4 Primitive reconstructed from first principles

Let

$$Z = (S_1, \dots, S_n, T)$$

have law p on a finite Cartesian-product alphabet. Fix a supported key

$$z = (s_1, \dots, s_n, t), \quad p(z) > 0.$$

For each nonempty collection $a \subseteq [n]$, define

$$E_a(s) = \bigcap_{i \in a} \{S_i = s_i\}.$$

For an antichain α of nonempty collections, define the inclusive disjunction

$$A_\alpha(s) = \bigcup_{a \in \alpha} E_a(s). \tag{1}$$

Every branch contains the keyed source realization. Therefore $p(A_\alpha) > 0$ and $p(T = t, A_\alpha) > 0$. The pointwise cumulative is

$$c_\alpha(z; p) = \log \frac{p(T = t, A_\alpha)}{p(T = t)p(A_\alpha)} = \log \frac{p(T = t \mid A_\alpha)}{p(T = t)}. \tag{2}$$

It is a local likelihood ratio for the *keyed* event, not yet an atom. The informative and misinformative components are

$$c_\alpha^+(z; p) = -\log p(A_\alpha), \quad c_\alpha^-(z; p) = \log \frac{p(T = t)}{p(T = t, A_\alpha)}, \quad (3)$$

and $c_\alpha = c_\alpha^+ - c_\alpha^-$. Both components are nonnegative; their difference can have either sign.

Finding F1 [P,R]. Equations (1)–(3) are defined for every supported finite key. They assign no pointwise value to a zero-probability realization.

4.1 Antichain order

Define

$$\alpha \preceq \beta \iff \forall b \in \beta \exists a \in \alpha : a \subseteq b. \quad (4)$$

Since $a \subseteq b$ implies $E_b(s) \subseteq E_a(s)$,

$$\alpha \preceq \beta \implies A_\beta(s) \subseteq A_\alpha(s). \quad (5)$$

Both c^+ and c^- increase along the lattice. Their difference need not.

4.2 Möbius atoms and exact reconstruction

Define atoms through the finite-poset zeta relation

$$c_\alpha^u(z; p) = \sum_{\beta \preceq \alpha} \pi_\beta^u(z; p), \quad u \in \{+, -, \text{net}\}, \quad (6)$$

where $\pi^{\text{net}} = \pi^+ - \pi^-$. Möbius inversion makes Equation (6) unique after the cumulatives are fixed. The greatest antichain is $\{[n]\}$ and its event is the complete source key, whence

$$c_{\{[n]\}}(z; p) = \log \frac{p(t \mid s_1, \dots, s_n)}{p(t)} = i(s_1, \dots, s_n; t). \quad (7)$$

Define averaged atoms

$$\Pi_\beta^u(p) = \sum_{z: p(z) > 0} p(z) \pi_\beta^u(z; p). \quad (8)$$

Linearity gives exact reconstruction:

$$\sum_\beta \Pi_\beta^{\text{net}}(p) = I(S_1, \dots, S_n; T). \quad (9)$$

Finding F2 [P,R]. This is an algebraic theorem about distribution-dependent cumulatives. It does not require atoms to be independent target coordinates.

4.3 What is actually nonnegative

Makkeh–Gutknecht–Wibral prove, for each supported key,

$$\pi_\beta^+(z; p) \geq 0, \quad \pi_\beta^-(z; p) \geq 0. \quad (10)$$

This does not imply that π_β^{net} or its average is nonnegative.

Independent rational checks found no violation of Equation (10): all 12,869 binary two-source empirical tables with $1 \leq N \leq 8$ (51,480 supported keys); all 4,844 binary three-source tables with $1 \leq N \leq 4$ (15,504 keys); and forty seeded binary four-source tables (1,060 keys, 166 atoms), with seed 1398296644. The first two searches were exhaustive within those bounds. Ratios were compared before logarithms. Bounded checks support but do not replace the proof.

Finding F3 [P,B]. No component-sign flaw was found. Universal nonnegativity of *net* atoms would be false.

5 What semantic choice is made?

For propositions $A_1, \dots, A_m$, a proposition C follows from every A_i if and only if it follows from their disjunction:

$$(\forall i, A_i \models C) \iff (A_1 \vee \dots \vee A_m) \models C. \quad (11)$$

The disjunction is, up to equivalence, the strongest proposition implied by every branch. Shared exclusions then identifies information shared by the source statements with the likelihood-ratio information supplied by that disjunction. This is coherent, but it is a constitutive semantic postulate, not a unique consequence of Shannon’s axioms. The parthood/logic isomorphism establishes an indexing structure; it does not force every scientifically useful redundancy measure to use Equation (2).

Finding F7 [P,R]. The logical claim is sound. Uniqueness of the statistical meaning is not established.

5.1 There is no single fixed keyed auxiliary variable

$A_\alpha(s)$ changes with the complete source key. Consequently the average in Equation (8) changes the event while weighting observations by $p(s, t)$. It is not generally $I(W; T)$ for one fixed random variable W . The 2021 paper states this, rejects two naive channel readings, and supplies a special metadata-masked channel interpretation. An application claiming one fixed “shared signal” needs an extra construction and equivalence proof.

5.2 Independent COPY has no common deterministic signal

Let S_1, S_2 be independent fair bits and $T = (S_1, S_2)$. Bottom Sx redundancy is

$$\Pi_{\{\{1\}, \{2\}\}}^{\text{net}} = \log \frac{4}{3} > 0. \quad (12)$$

If $U = f_1(S_1) = f_2(S_2)$ almost surely, then $f_1(S_1)$ and $f_2(S_2)$ are independent and equal. Thus U is independent of itself. For every u ,

$$\Pr(U = u) = \Pr(U = u)^2,$$

so every mass is zero or one and U is constant. Therefore $I(U; T) = 0$ while Equation (12) is positive.

Finding F8 [X]. Sx redundancy is not Gács–Körner common information or the entropy of a nonconstant statistic computable from each independent source. Its commonality is keyed and event-logical.

6 Exact axiomatic trade-offs

6.1 Re-encoding invariance and target chain rule

Coordinatewise bijections preserve equality events and all probabilities in Equations (1)–(3). They preserve every cumulative and atom up to source-label permutation. A joint invertible map mixing sources changes the source partition and is not covered.

For $T = (T_1, T_2)$ and one fixed keyed event A ,

$$\log \frac{p(t_1, t_2 \mid A)}{p(t_1, t_2)} = \log \frac{p(t_1 \mid A)}{p(t_1)} + \log \frac{p(t_2 \mid t_1, A)}{p(t_2 \mid t_1)}. \quad (13)$$

The identity propagates through linear inversion and averaging, using the conditional law in the second term.

6.2 Exact identity and local-positivity failures

For independent COPY, every key has $p(A) = 3/4$, $p(t) = 1/4$, and $p(t, A) = 1/4$. The complete atom vector is

$$(R, U_1, U_2, \text{Syn}) = \left(\log \frac{4}{3}, \log \frac{3}{2}, \log \frac{3}{2}, \log \frac{4}{3} \right), \quad (14)$$

and sums to $\log 4$. Identity redundancy would instead require $R = I(S_1; S_2) = 0$.

With the same sources but $T = S_2$, direct evaluation gives

$$(R, U_1, U_2, \text{Syn}) = \left(\log \frac{4}{3}, -\log \frac{4}{3}, \log \frac{3}{2}, \log \frac{4}{3} \right). \quad (15)$$

The negative unique atom has exact split

$$U_1^+ = \log \frac{3}{2}, \quad U_1^- = \log 2, \quad U_1 = -\log \frac{4}{3}. \quad (16)$$

6.3 What arXiv:2512.16662v1 implies

Matthias, Makkeh, Wibral, and Gutknecht prove that no general discrete antichain PID satisfies all of net-atom local positivity (LP), target chain rule (TCR), and coordinatewise invertible re-encoding invariance (REI). They also prove a related LP–identity–REI incompatibility. SxPID has REI and TCR, but not LP or identity. Equations (14) and (15) place it exactly on that trade-off.

This is a structural incompatibility, not an algebraic inconsistency of SxPID. It does not imply that Sx fails reconstruction. Version 1, dated 18 December 2025, is the exact version audited; arXiv still listed it as current when rechecked on 25 July 2026.

Finding F6 [P,R,X,E]. Sx preserves TCR and REI by accepting signed net atoms. A claim that it also has universal net local positivity is false.

6.4 XOR-source-copy executable witness

Let S_1, S_2 be independent fair bits, $S_3 = S_1 \oplus S_2$, and $T = (S_1, S_2, S_3)$. The current kernel gives

$$\Pi_{\{\{1\}, \{2\}\}} = \Pi_{\{\{1\}, \{3\}\}} = \Pi_{\{\{2\}, \{3\}\}} = \log \frac{4}{3}, \quad (17)$$

$$\Pi_{\{\{1\}, \{23\}\}} = \Pi_{\{\{2\}, \{13\}\}} = \Pi_{\{\{3\}, \{12\}\}} = \log \frac{9}{8}, \quad (18)$$

$$\Pi_{\{\{12\}, \{13\}, \{23\}\}} = \log \frac{32}{27}. \quad (19)$$

All other atoms vanish, and

$$\left(\frac{4}{3} \right)^3 \left(\frac{9}{8} \right)^3 \frac{32}{27} = 4,$$

so the atoms sum to $2 \log 2$. This one unconditional witness need not display a negative atom; the 2025 result concerns universal property combinations.

7 Exact counterexample ledger

7.1 Negative redundancy and pairwise non-identifiability

For independent fair S_1, S_2 and XOR target $T = S_1 \oplus S_2$, every key has $p(A) = 3/4$, $p(t) = 1/2$, and $p(t, A) = 1/4$. Thus

$$R_{\text{sx}} = \log \frac{2}{3} < 0. \quad (20)$$

The vector

$$\left(\log \frac{2}{3}, \log \frac{3}{2}, \log \frac{3}{2}, \log \frac{4}{3} \right)$$

sums to $\log 2$. Compare this law with three independent fair bits S_1, S_2, T . Both have the same independent pairwise marginals (S_1, T) and (S_2, T) , but

$$R_{\text{sx}}^{\text{XOR}} = \log \frac{2}{3}, \quad R_{\text{sx}}^{\text{independent}} = 0. \quad (21)$$

The full joint law, not pairwise marginals, identifies SxPID.

7.2 An irrelevant source and source–target role asymmetry

Add an independent fair S_3 to XOR without changing the target. The bottom event has $p(A) = 7/8$, $p(t, A) = 3/8$, hence

$$R_{\text{sx}}(S_1, S_2, S_3 \rightarrow T) = \log \frac{6}{7}, \quad (22)$$

which is larger than $\log(2/3)$. Naive net source monotonicity is false.

For a separate exact role witness, assign probability $1/3$ to each row

$$(S_1, S_2, T) \in \{(0, 1, 1), (1, 0, 0), (1, 1, 1)\}.$$

Then

$$R_{\text{sx}}(S_1, S_2 \rightarrow T) = \frac{1}{3} \log \frac{9}{4}, \quad (23)$$

$$R_{\text{sx}}(T, S_2 \rightarrow S_1) = \frac{1}{3} \log \frac{27}{16}. \quad (24)$$

Their difference is $\frac{1}{3} \log(4/3) > 0$. PID is target-directed even though ordinary mutual information is symmetric.

7.3 No universal atomwise data processing

Starting from independent COPY, coarsen the target deterministically to XOR and then to a constant. Bottom redundancy follows

$$\log \frac{4}{3} \longrightarrow \log \frac{2}{3} \longrightarrow 0. \quad (25)$$

The first step changes sign; the second removes all target information but raises signed redundancy from a negative number to zero. Coarsening one XOR source to a constant also changes $\log(2/3)$ to zero. Total mutual information obeys data processing; individual signed Sx atoms do not inherit a universal DPI. Preprocessing changes the estimand.

8 The 2026 structural-impossibility challenge

8.1 Exact witness pair

Lyu, Clark, and Raviv (arXiv:2604.03869v2, 14 April 2026) define two systems. For the *hat* system let

$$x_1, x_2, x_4, x_5, x_7, x_8 \stackrel{\text{iid}}{\sim} \text{Bernoulli}(1/2),$$

with

$$x_3 = x_1 \oplus x_2, \quad x_6 = x_4 \oplus x_5, \quad x_9 = x_7 \oplus x_8.$$

For the *tilde* system let

$$x_1, x_2, x_4, x_5, x_7 \stackrel{\text{iid}}{\sim} \text{Bernoulli}(1/2),$$

with

$$x_3 = x_1 \oplus x_2, \quad x_6 = x_4 \oplus x_5, \quad x_9 = x_1 \oplus x_5, \quad x_8 = x_7 \oplus x_1 \oplus x_5.$$

Both use

$$S_1 = (x_1, x_4, x_7), \quad S_2 = (x_2, x_5, x_8), \quad S_3 = (x_3, x_6, x_9), \quad T = (x_1, x_5, x_9). \quad (26)$$

The hat system has 64 equiprobable states and three independent target bits, so $I(\hat{S}; \hat{T}) = 3 \log 2$. The tilde system has 32 states and $x_9 = x_1 \oplus x_5$, so $I(\tilde{S}; \tilde{T}) = 2 \log 2$.

8.2 What Definition 6 and Lemma 4 assign

Each target coordinate has the same minimal recovering antichain in both systems:

$$\alpha(x_1) = \{\{1\}, \{23\}\}, \quad \alpha(x_5) = \{\{2\}, \{13\}\}, \quad \alpha(x_9) = \{\{3\}, \{12\}\}. \quad (27)$$

Their Lemma 4 stipulates

$$\Pi(\alpha) = H(U_\alpha). \quad (28)$$

It assigns one bit to each of those labels and zero elsewhere in both systems. In the tilde system the three coordinates are related. Their stipulated values sum to three bits while the target has two. This is a recoverability descriptor vector, not a whole-equals-sum-of-parts PID of $I(S; T)$ there.

The proof assigns $H(U_\alpha)$ directly; it does not invert an Sx cumulative, and it does not prove independence among distinct U_α . Theorem 1 is valid on its explicitly stated Definition 6 domain: the witness laws have the same complete descriptor vector and unequal mutual information, so no single map from that vector reconstructs mutual information for every law in that domain. Access labels and marginal component entropies omit the witness's cross-coordinate relation.

8.3 Exact SxPID result

Independent exact event counting and the current Rust kernel instead give eight nonzero net atoms.

Antichain family	Hat atom	Tilde atom
$\{\{1\}, \{2\}, \{3\}\}$	$\log(16/11)$	$\log(8/5)$
Each of $\{\{1\}, \{2\}\}, \{\{1\}, \{3\}\}, \{\{2\}, \{3\}\}$	$\log(11/10)$	$\log(15/14)$
Each label in Equation (27)	$\log(25/22)$	$\log(49/45)$
$\{\{12\}, \{13\}, \{23\}\}$	$\log(352/125)$	$\log(540/343)$

The other ten atoms vanish. Exact product identities establish reconstruction before evaluating logarithms:

$$\frac{16}{11} \left(\frac{11}{10}\right)^3 \left(\frac{25}{22}\right)^3 \frac{352}{125} = 8, \quad (29)$$

$$\frac{8}{5} \left(\frac{15}{14}\right)^3 \left(\frac{49}{45}\right)^3 \frac{540}{343} = 4. \quad (30)$$

Taking logs gives $3 \log 2$ and $2 \log 2$. Eight of eighteen net atoms differ; ten agree exactly. The largest difference is

$$\log \frac{352/125}{540/343} \approx 0.58147874590342 \text{ nats} = 0.83889650309719 \text{ bits}. \quad (31)$$

At each recovery label, the exact component atoms are

$$\hat{\Pi}^+ = \log \frac{225}{176}, \quad \hat{\Pi}^- = \log \frac{9}{8}, \quad \tilde{\Pi}^+ = \log \frac{49}{40}, \quad \tilde{\Pi}^- = \log \frac{9}{8}.$$

The target relation changes informative increments although the access labels stay fixed.

8.4 What arXiv:2604.03869v2 does and does not imply

Valid theorem on the stated Definition 6 domain [P,R]. The exhibited laws have equal complete vectors $(H(U_\alpha))_\alpha$ and unequal mutual information. Therefore no single function of that descriptor vector reconstructs mutual information for every law in that domain.

Not established [X,E]. The paper does not establish that the two laws have equal SxPID atoms or that SxPID fails MI reconstruction. Equations (29) and (30) show the opposite.

8.4.1 Descriptor-factorization firewall

The transfer condition does not require PID terminology. Let $D(P)$ be a descriptor vector, $A(P)$ an atom vector, and $J(P)$ the quantity to be reconstructed. Suppose the atom assignment factors through the descriptor:

$$A = g \circ D. \quad (32)$$

If two systems satisfy

$$D(P) = D(Q), \quad J(P) \neq J(Q), \quad (33)$$

then $A(P) = A(Q)$, so no function f can satisfy $f(A(R)) = J(R)$ for every system R . Conversely,

$$D(P) = D(Q), \quad A(P) \neq A(Q) \implies \nexists g \ A = g \circ D. \quad (34)$$

This is the exact logical firewall. The Lyu–Clark–Raviv witness proves a reconstruction impossibility for atom assignments that accept Equation (32), using their recoverability-label and component-entropy descriptor. It does not transfer merely because another assignment is indexed by the same antichain lattice. A separate theorem must establish Equation (32) for that assignment.

The SxPID calculation supplies the converse witness in Equation (34): the descriptors agree but eight Sx atoms differ. Therefore SxPID does not factor through that descriptor map. The descriptor theorem remains valid; its operative premise is descriptor sufficiency, not antichain indexing alone.

The generic firewall is kernel-checked in `audit/formal/lean-foundational-sxpid/PidDescriptorFactorization.lean`. Its three theorems have an empty Lean axiom inventory. The pinned checker `scripts/check-lean-descriptor-factorization.py` and fail-closed three-mutation suite `scripts/check-lean-descriptor-factorization-self-test.py` bind the result to the recorded source, Lake manifest/configuration, toolchain identifier, and reported Lean version. The latter also kernel-checks three finite premise countermodels. Their deterministic outputs are `audit/evidence/foundational-sxpid-descriptor-factorization-lean.json` and `audit/evidence/foundational-sxpid-descriptor-factorization-mutations.json`. These are reproducibility checks, not an authenticity or binary-attestation claim about the installed Lean/mathlib toolchain.

Phrases such as “regardless of any axioms” and “any multivariate information decomposition that relies solely on the antichain lattice” are therefore safe only with the descriptor-factorization premise explicit. Distribution-dependent signed Möbius atoms, including SxPID, do not accept that premise in this witness.

Preserved insight. Access labels alone do not encode every relation among semantically named target components. Relation-aware metadata may be useful, but it cannot replace Sx atoms with nonadditive coordinate entropies and remain a decomposition of mutual information.

Finding F11 [P,R,X,E]. The witness is a genuine warning against access-label/marginal-entropy descriptors as a complete representation. It is not a contradiction of shared-exclusions PID.

8.5 Three independent executable and formal routes

The standard-library checker `audit/tools/foundational_sxpid/check_lcr_relation_witness.py` does not import `pid-rs`. It:

1. verifies the generated rows match the paper’s fair-bit/XOR constructions and source/target index sets;
2. checks Definition 6 conditions (i), (ii), and (iii) exactly on both finite laws, including mutual rather than pairwise within-source independence and all twenty-four target-component/source-group cases per system;
3. derives all minimal recovering antichains and both complete Lemma 4 descriptor vectors from the rows, proving those vectors equal;
4. generates all eighteen nonempty antichains of the seven nonempty three-source subsets;
5. independently implements Equation (4);
6. counts all keyed equality-event unions and target intersections;
7. performs exact Möbius inversion in formal prime-exponent logarithms;
8. verifies all local component atoms are constant across these two equiprobable supports;
9. derives the eight differing atoms, ten equal atoms, and products 8 and 4; and
10. kills union-to-intersection, row-derived Lemma 4 descriptor substitution, and cross-relation mutations.

Its full output is `audit/evidence/foundational-sxpid-lcr-exact-audit.json`, bound to the checker, generated rows, Rust regression, and Rust kernel by SHA-256 digests. The independent Rust route is `crates/pid-core/tests/sxpid_relation_witness.rs`; it calls the public categorical kernel, checks every displayed atom and component, enforces the integer product identities, and rejects descriptor substitution.

The third route is the generic Lean factorization firewall above. Its proof has no axioms in the kernel inventory. The self-test kills removal of the factorization premise, replacement of the unequal-quantity premise by equality, and replacement of the unequal-atom premise by equality. Three finite Bool/Unit countermodels show why the altered premises cannot support the original conclusions: equal descriptors alone need not equalize atoms, equal quantities can admit reconstruction, and equal atoms can admit factorization. This route proves the transfer logic, not the concrete Sx event calculation; the exact-rational and Rust routes supply that separate binding.

At generation, the bound SHA-256 values were:

- checker: 8244bd6a3a3a590f8a8da9f63ce5d5114d3453cb4623bde70ce02c7ddbeec96a;
- Rust regression: e11076dfd0bc8e8b3f3565128ed87c2cabe03c0f25b966e76fe925a09e3451ed;
- Rust kernel: f08b56bac473474b39dd2cf2f09c5d13ccda025825a011ee1870f6f2979ff98a.

9 Support, continuity, and estimation

Every probability in Equation (2) is positive at a supported key. If that key disappears, its local quantity has no operational occurrence; along rare-key sequences a local term can diverge like $-\log p(z)$. The 2021 differentiability theorem is an interior theorem on a fixed finite simplex, not pointwise continuity at disappearing keys.

A separate project theorem proves that support-restricted *averaged* cumulatives and full-lattice atoms on one fixed finite alphabet extend continuously through support changes. A

vanishing weight can control the local log divergence:

$$\text{local value may diverge} \quad \text{while} \quad p(z)\pi_\alpha(z;p) \longrightarrow 0.$$

This does not make absent-key pointwise quantities continuous.

For a known finite alphabet, the full joint population law identifies SxPID. Plug-in consistency does not supply support discovery, useful sparse-law rates, protection against adaptive quantization, generic post-selection/dependence inference, binary64 sign certificates, or causal identification. Reports need the estimand and source partition, alphabet and occupancy, sampling contract, preprocessing split, both components, sign status, reconstruction residual, sensitivity analysis, and an explicit noncausal boundary.

Finding F4/F10 [R]. Exact-real continuity and fixed-alphabet consistency are valid stability results; they do not establish practical calibration in every data regime.

10 Interpretive and causal limits

A net atom is a Möbius increment in a likelihood ratio. Its sign compares informative and misinformative increments. Without additional assumptions it does not mean beneficial mechanism, causal redundancy/synergy, correct sensor, deception, intent, replaceability, intervention robustness, safety margin, or decision correctness.

SxPID is a functional of an observational joint law. Distinct causal graphs can induce the same law and SxPID. No statistic of that law can distinguish them absent causal assumptions or interventions. The published operational interpretation also assumes a particular metadata-masked event channel. A deployment needs a loss model, calibration, sequential error policy, drift and out-of-distribution rules, and abstention. SxPID can be evidence, not an authorization rule.

Finding F12 [R]. Causal or authority-grade conclusions require a separate theorem connecting the observational Sx estimand to the needed intervention or decision quantity.

11 Validity-regime contract

Categorical SxPID is defensible when variables are categorical (or quantization is explicitly the estimand), the scientific source partition is fixed, the question is shared logical-exclusion information, signed atoms are acceptable, both components are retained, the full joint law is estimable under a declared sampling regime, dependence/drift/selection are separately controlled, cancellation and reconstruction are checked, and interpretation remains observational absent a causal bridge.

Another object is required for a common deterministic random variable, only nonnegative parts, identity redundancy, arbitrary local-processing monotonicity, source–target symmetry, pairwise identification, causal effect, unsupported high-dimensional continuous PID, or an authority-grade probability of correctness. Those are different mathematical requirements.

12 Formal-verification consequences

12.1 Exact specification and executable refinement

Formalize finite alphabets; supported keys; equality events; collections and antichains; Equations (1)–(3); the order and containment; concrete Möbius inversion; the top-node identity; averaged reconstruction; component positivity; coordinatewise REI; and target chain. Formalize

identity, LP, pairwise-identification, role-symmetry, irrelevant-source, and net-DPI counterexamples as theorems. These negative specifications prevent silent future claim strengthening.

Prove that canonical row counting equals the empirical law; each event count equals its predicate; the Rust order matches Equation (4); every cumulative reconstructs; specialized/general paths agree; admitted integer count products cannot overflow; and reports preserve signed components.

12.2 Certified numerical and semantic layers

Counts give rational pre-log quantities. Directed-rounding intervals can certify every log, atom, sum, and reconstruction. Report certified positive, certified negative, or unresolved sign rather than clamping.

Keep semantic boundaries explicit: disjunction as shared logical consequence is an accepted postulate; uniqueness among all redundancy meanings is not a theorem; atoms are not common deterministic coordinates or nonnegative set sizes; observational atoms are not causal effects. Formal methods can prove a precise model correct, but cannot make an unchosen semantics uniquely true.

13 Correctness-preserving research directions

1. **Relation annotations, not replacement atoms [O]**. Attach target multi-information, algebraic constraints, or declared factor graphs as metadata while preserving Equation (9). Do not add them again as atoms without a no-double-counting theorem.
2. **Semantic uniqueness class [O]**. Find the weakest logical, epistemic, decision-theoretic, or Shannon axioms under which disjunction is uniquely selected.
3. **Coarse-graining calculus [O]**. Classify maps that preserve the keyed event sigma-algebra, merge atom families in a known way, or preserve signs. Keep Equations (22) and (25) as negative controls.
4. **Operational channel validation [O]**. Test whether real observers receive the assumed metadata-masked channel. Leakage or learning across uses changes the operational estimand.
5. **Finite-sample signed certification [O]**. Combine valid concentration, closed-simplex continuity, and intervals while separating model, sampling, selection, and numerical error.

14 Final claim ledger

Claim	Decision	Evidence
Keyed cumulative is a valid local likelihood ratio	Established	[P,R], Equations (1)–(3)
Atoms reconstruct mutual information	Established	[P,R,E], Equations (6)–(9)
Component atoms are nonnegative	Established/ bounded-tested	[P,B]
Net atoms are nonnegative	False	[X,E], Equations (15), (20)
Coordinatewise REI and target chain hold	Established	[P,R]
Identity holds	False	[X,E], Equation (14)
Sx redundancy is common deterministic information	False	[X]

Claim	Decision	Evidence
Pairwise marginals identify Sx redundancy	False	[X], Equation (21)
Net atoms obey arbitrary processing DPI	False	[X], Equation (25)
Source/target roles are interchangeable	False	[X], Equations (23)–(24)
Disjunction semantics is logically coherent	Established	[P,R]
Disjunction is uniquely forced by Shannon theory	Not established	Constitutive-semantic gap
The Lyu systems have identical SxPID atoms	False	[X,E], Equations (29)–(30)
Lyu descriptors expose lost target relations	Established in their model	[P,R]
Descriptor-collision impossibility transfers to every antichain PID	False without a factorization theorem	[P,R,X,E], Equations (32)–(34)
Lyu refutes all lattice PID including Sx	Not established for Sx	Descriptor factorization fails and three routes agree
SxPID is causal/authority-grade by itself	False/unsupported	Observational identification boundary
Categorical SxPID is a defensible signed event-logical estimand	Yes under the regime contract	Entire audit

15 Reproducibility record

The Rust witness was evaluated against commit 6e29b3f5badc337a58eb821babc41641037e15d0. The audited kernel SHA-256 was

f08b56bac473474b39dd2cf2f09c5d13ccda025825a011ee1870f6f2979ff98a.

The environment was Rust/Cargo 1.96.0 and Python 3.14.6. The permanent commands are

```
python3 -I -S audit/tools/foundational_sxpid/check_lcr_relation_witness.py \
  --write-evidence audit/evidence/foundational-sxpid-lcr-exact-audit.json
cargo test --locked -p pid-core --test sxpid_relation_witness
python3 -I -S scripts/check-lean-descriptor-factorization.py
python3 -I -S scripts/check-lean-descriptor-factorization-self-test.py
```

At creation both Rust tests, the independent exact checker, and the Lean factorization checks passed. The rational/formal-log route derives event counts, order, atoms, and mutations without invoking the Rust implementation. The Rust route independently exercises the public event/lattice kernel. The Lean route proves the abstract transfer firewall while explicitly leaving the concrete descriptor/Sx binding to the two executable routes.

Primary sources

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2. A. J. Gutknecht, M. Wibral, and A. Makkeh, “Bits and Pieces: Understanding Information Decomposition from Part-whole Relationships and Formal Logic,” *Proceedings of the Royal Society A* 477, 20210110 (2021), doi:10.1098/rspa.2021.0110, arXiv:2008.09535.

3. P. H. Matthias, A. Makkeh, M. Wibral, and A. J. Gutknecht, “Novel Inconsistency Results for Partial Information Decomposition,” [arXiv:2512.16662v1](#), 18 December 2025. Version 1 was audited and remained current on arXiv when rechecked on 25 July 2026.
4. A. Lyu, A. Clark, and N. Raviv, “Structural Impossibility of Antichain-Lattice Partial Information Decomposition,” [arXiv:2604.03869v2](#), 14 April 2026.
5. P. L. Williams and R. D. Beer, “Nonnegative Decomposition of Multivariate Information,” [arXiv:1004.2515](#) (2010).